

Shopping Cart Calculator using HTML, CSS and JavaScript

Experiment Report — Arrays, Objects and DOM Manipulation

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Aim

To design a Shopping Cart Calculator web page using HTML and CSS, and to implement its logic in JavaScript using arrays of objects, dynamic DOM element creation, and discount calculation.

Software / Tools Required

- Visual Studio Code
- Google Chrome (or any modern web browser)
- HTML5
- CSS3
- JavaScript

Program Code

The cart stores items as an array of objects (**cartItems**), each with a name and price. **addSelectedItem()** adds a predefined product chosen from a dropdown, while **addCustomItem()** validates and adds a user-entered product. **displayCart()** rebuilds the list dynamically using `createElement`, and **calculateDiscount()** computes the subtotal, discount amount, and final total. An inline internal script defines **clearCart()** to reset the cart.

Note: The original script.js contained a stray character ('c') on its own line before the `cartItems` declaration, which would cause a `ReferenceError` in the browser. This has been corrected in the listing below, and a footer with the student's name and PRN has been added to `index.html` and `style.css`.

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Shopping Cart Calculator</title>

<link rel="stylesheet" href="style.css">

</head>
<body>

<div class="container">

  <h1>■ Shopping Cart Calculator</h1>

  <div class="box">

    <h3>Select Product</h3>

    <select id="product">
      <option value="60"> Milk - ■60</option>
      <option value="250"> Mangoes - ■250</option>
```

```

        <option value="120"> Chocolate - ■120</option>
        <option value="2500"> Whey Protein - ■2500</option>
    </select>

    <button onclick="addSelectedItem()">Add Selected Item</button>
</div>

<hr>

<div class="box">

    <h3>Add Custom Product</h3>

    <input type="text" id="item" placeholder="Product Name">

    <input type="number" id="price" placeholder="Price">

    <button onclick="addCustomItem()">Add Custom Item</button>
</div>

<hr>

<div class="box">

    <h3>Discount</h3>

    <input
    type="number"
    id="discount"
    value="0"
    min="0"
    max="100"
    onchange="calculateDiscount()">

</div>

<hr>

<h2>Cart Items</h2>

<ul id="cart"></ul>

<div class="bill">

    <h3>Subtotal : ■ <span id="subtotal">0</span></h3>

    <h3>Discount : ■ <span id="discountAmount">0</span></h3>

    <h2>Total : ■ <span id="finalTotal">0</span></h2>

</div>

<button class="clear" onclick="clearCart()">Clear Cart</button>

</div>

<!-- Footer -->
<footer>
    <p>Developed by <strong>Parth Denge</strong></p>
    <p>PRN: <strong>24070521018</strong></p>
</footer>

```

```
<script>

function clearCart()
{
    cartItems=[];
    displayCart();
}

</script>

<script src="script.js"></script>

</body>
</html>
```

style.css

```
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial, Helvetica, sans-serif;
}

body{

    background:linear-gradient(135deg,#1d2671,#c33764);

    display:flex;

    justify-content:center;

    align-items:center;

    min-height:100vh;
}

.container{

    width:500px;

    background:white;

    padding:30px;

    border-radius:20px;

    box-shadow:0 15px 35px rgba(0,0,0,.4);
}

h1{

    text-align:center;

    color:#222;

    margin-bottom:20px;
}

.box{

    margin-bottom:20px;
```

```
}

h3{

    margin-bottom:10px;

    color:#444;

}

select,
input{

    width:100%;

    padding:12px;

    margin-bottom:10px;

    border-radius:8px;

    border:1px solid #bbb;

    font-size:16px;

}

button{

    width:100%;

    padding:12px;

    border:none;

    border-radius:8px;

    background:#2196f3;

    color:white;

    font-size:16px;

    cursor:pointer;

    transition:.3s;

}

button:hover{

    background:#0d6efd;

    transform:scale(1.03);

}

.clear{

    background:#945755;

    margin-top:20px;

}

.clear:hover{
```

```
        background:#c62828;
    }
    ul{
        list-style:none;
        margin-top:15px;
    }
    li{
        display:flex;
        justify-content:space-between;
        background:#f4f4f4;
        margin:8px 0;
        padding:12px;
        border-left:5px solid #2196f3;
        border-radius:8px;
        font-size:17px;
    }
    .bill{
        margin-top:20px;
        background:#fafafa;
        padding:15px;
        border-radius:10px;
        box-shadow:0 0 8px rgba(95, 4, 4, 0.1);
    }
    .bill h2{
        color:rgb(91, 247, 91);
        margin-top:10px;
    }
    .bill h3{
        margin-top:8px;
    }
    hr{
        margin:20px 0;
    }
    footer{
```

```
text-align:center;

margin-top:15px;

color:white;

font-size:14px;

}
```

script.js

```
let cartItems=[];


function addSelectedItem()
{

    let product=document.getElementById("product");

    let name=product.options[product.selectedIndex].text.split("-")[0].trim();

    let price=Number(product.value);

    cartItems.push({

        name:name,

        price:price

    });

    displayCart();

}


function addCustomItem()
{

    let item=document.getElementById("item").value;

    let price=Number(document.getElementById("price").value);

    if(item=="" || price<=0)
    {
        alert("Enter valid details.");
        return;
    }

    cartItems.push({

        name:item,

        price:price

    });

    document.getElementById("item").value="";

    document.getElementById("price").value="";

}
```

```
    displayCart();  
}
```

```
function displayCart()  
{  
    let cart=document.getElementById("cart");  
    cart.innerHTML="";  
    let subtotal=0;  
    for(let i=0;i<cartItems.length;i++)  
    {  
        subtotal+=cartItems[i].price;  
        let li=document.createElement("li");  
        li.innerHTML=  
            "<span>" + cartItems[i].name + "</span>" +  
            "<span>■" + cartItems[i].price + "</span>";  
        cart.appendChild(li);  
    }  
    document.getElementById("subtotal").innerHTML=subtotal;  
    calculateDiscount();  
}
```

```
function calculateDiscount()  
{  
    let subtotal=0;  
    for(let i=0;i<cartItems.length;i++)  
    {  
        subtotal+=cartItems[i].price;  
    }  
    let discount=Number(document.getElementById("discount").value);  
    if(discount<0)  
        discount=0;  
    if(discount>100)  
        discount=100;  
    let discountAmount=subtotal*discount/100;
```

```
    let total=subtotal-discountAmount;

    document.getElementById("discountAmount").innerHTML=discountAmount.toFixed(2);

    document.getElementById("finalTotal").innerHTML=total.toFixed(2);

}
```

```
displayCart();
```

Output

Attach a screenshot of the running program. The screenshot must clearly show:

- Student Name: Parth Denge
- PRN: 24070521018
- File Path
- Cart items added, subtotal, discount, and final total.

[Insert Screenshot Here]

Result / Conclusion

Result:

The Shopping Cart Calculator was developed successfully using HTML, CSS, and JavaScript. The program allows adding predefined and custom products to the cart, dynamically updates the cart list, and correctly computes the subtotal, discount amount, and final payable total.

Conclusion:

This experiment demonstrates the use of arrays of objects to model cart data, dynamic DOM element creation with `createElement` and `appendChild`, event-driven updates (`onclick`, `onchange`), and arithmetic operations for computing discounts and totals in a real-world style application.